UNIT – II

CONTROL FLOW, LOOPS

Conditionals: Boolean values and operators, conditional (if), alternative (if-else), chained conditional (if-elif-else); Iteration: while, for, break, continue.

Control Flow, Loops:

Boolean Values and Operators:

A boolean expression is an expression that is either true or false. The following examples use the operator ==, which compares two operands and produces True if they are equal and False otherwise:

>>> 5 == 5

True

>>> 5 == 6

False

True and False are special values that belong to the type bool; they are not strings:

>>> type(True)

<class 'bool'>

>>> type(False)

<class 'bool'>

The == operator is one of the relational operators; the others are: x != y # x is not equal to y

x > y # x is greater than y x < y # x is less than y

x >= y # x is greater than or equal to y x <= y # x is less than or equal to y

Note:

All expressions involving relational and logical operators will evaluate to either true or false

Conditional (if):

The if statement contains a logical expression using which data is compared and a decision is made based on the result of the comparison.

Syntax:

```
if expression:
    statement(s)
```

If the boolean expression evaluates to TRUE, then the block of statement(s) inside the if statement is executed. If boolean expression evaluates to FALSE, then the first set of code after the end of the if statement(s) is executed.

if Statement Flowchart:

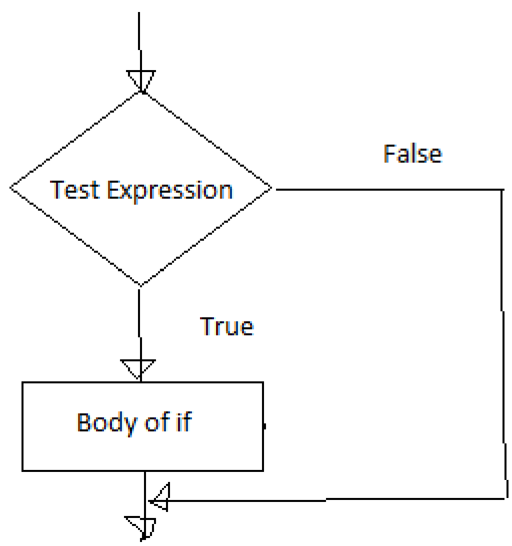

Fig: Operation of if statement

Example: Python if Statement

```
a = 3
if a > 2:
    print(a, "is greater")
print("done")

a = -1
if a < 0:
    print(a, "a is smaller")
print("Finish")
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/if1.py
3 is greater
done
-1 a is smaller
Finish

```
a=10

if a>9:

    print("A is Greater than 9")
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/if2.py

A is Greater than 9

Alternative if (If-Else):

An else statement can be combined with an if statement. An else statement contains the block of code (false block) that executes if the conditional expression in the if statement resolves to 0 or a FALSE value.

The else statement is an optional statement and there could be at most only one else Statement following if.

Syntax of if - else :

```
if test expression:
        Body of if stmts
else:
        Body of else stmts
```

If - else Flowchart :

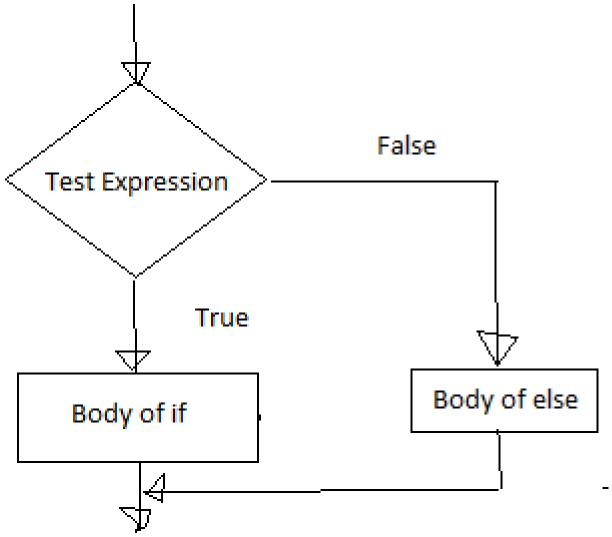

Fig: Operation of if – else statement

Example of if - else:

```
a=int(input('enter the number'))
if a>5:
    print("a is greater")
else:
    print("a is smaller than the input given")
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/ifelse.py
enter the number 2
a is smaller than the input given

```
a=10
b=20
if a>b:
    print("A is Greater than B")
else:
    print("B is Greater than A")
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/if2.py
B is Greater than A

Chained Conditional: (If-elif-else):

The elif statement allows us to check multiple expressions for TRUE and execute a block of code as soon as one of the conditions evaluates to TRUE. Similar to the else, the elif statement is optional. However, unlike else, for which there can be at most one statement, there can be an arbitrary number of elif statements following an if.

Syntax of if – elif - else :

```
If test expression:
        Body of if stmts
elif test expression:
        Body of elif stmts
else:
        Body of else stmts
```

Flowchart of if – elif - else:

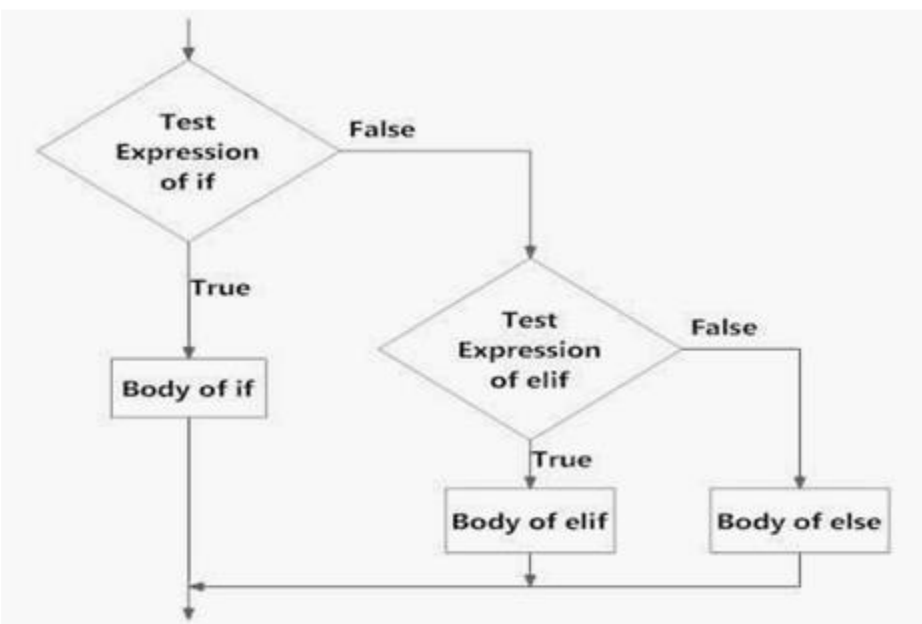

Fig: Operation of if – elif - else statement

Example of if - elif – else:

```
a=int(input('enter the number'))
b=int(input('enter the number'))
c=int(input('enter the number'))
if a>b:
```

```
    print("a is greater")
elif b>c:
    print("b is greater")
else:
    print("c is greater")
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/ifelse.py
enter the number5
enter the number2
enter the number9
a is greater
>>>
C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/ifelse.py
enter the number2
enter the number5
enter the number9
c is greater

```
var = 100
if var == 200:
    print("1 - Got a true expression value")
    print(var)
elif var == 150:
    print("2 - Got a true expression value")
    print(var)
elif var == 100:
    print("3 - Got a true expression value")
    print(var)
else:
    print("4 - Got a false expression value")
    print(var)
print("Good bye!")
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/ifelif.py

3 - Got a true expression value

100

Good bye!